

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		6.45 cm ³		1		1		
		(ii)		moles thiosulfate = $0.500 \times \frac{6.45}{1000} = 3.225 \times 10^{-3}$ mol (1) reaction ratio $\Rightarrow 2\text{S}_2\text{O}_3^{2-} \equiv 1\text{I}_2 \equiv 1\text{ClO}^-$ so moles chlorate(I) = 1.6125×10^{-3} mol (1) concentration chlorate(I) = $\frac{1.6125 \times 10^{-3}}{25 \times 10^{-3}} \times 10 = 0.645$ mol dm ⁻³ (1) ecf possible from part (i)		3		3	2	
		(iii)		$0.645 \times \frac{74.5}{10} = 4.81\%$		1		1	1	
		(iv)		should have chosen 0.200 mol dm ⁻³ marks credited for reasons, MAX 1 mark for reasons if a different concentration is chosen use a lower concentration so titration volume is greater – smaller percentage error / more accurate (1) cannot use too low a concentration / 0.0500 mol dm ⁻³ as volume would be too large (for a standard burette) (1)			2	2		2
	(b)	(i)	1	order with respect to $\text{ClO}_3^- \Rightarrow$ first order (1) order with respect to $\text{Br}^- \Rightarrow$ first order (1) must show working to gain each mark		2		2	2	